

SAO JOSE DOS CAMPOS, Brazil

WMO: 838093

Lat: 23.229S

Lon: 45.862W

Elev: 646

StdP: 93.80

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	10.1	4.1	5.5	21.2	6.0	6.3	19.7	7.5	20.3	6.7	20.3	0.6	180	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.3	34.0	20.7	32.9	20.9	31.9	21.0	24.0	29.0	23.5	28.3	23.0	27.8	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.2	25.0	22.2	18.2	24.4	21.9	17.9	24.3	75.8	28.8	73.5	28.7	71.5	27.4	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.0	5.8	DB	5.9	36.1	2.5	1.5	4.2	37.2	2.7	38.1	1.4	38.9	-0.4	40.0	
			WB	5.4	25.6	2.4	1.7	3.7	26.8	2.3	27.8	0.9	28.7	-0.8	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	24.8	25.3	24.4	22.3	19.6	18.2	18.0	19.3	20.9	22.4	22.8	24.2
	DBStd	3.47	2.16	1.81	1.78	2.27	2.33	2.43	2.70	2.66	2.99	2.87	2.57	2.16
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	121	0	0	0	1	14	31	38	21	10	3	2	0
	CDD10.0	4313	458	428	446	369	297	246	247	289	328	383	383	439
	CDD18.3	1393	200	194	188	120	53	26	27	52	88	128	134	181
	CDH23.3	9346	1300	1336	1135	683	279	189	258	508	733	891	850	1184
Wind	WSAvg	2.0	1.9	1.9	1.8	1.8	1.6	1.4	1.5	1.9	2.4	2.5	2.5	2.3
	PrecAvg	1374	236	208	184	67	57	34	33	29	83	124	142	172
	PrecMax	1808	386	450	340	154	107	140	94	64	165	245	247	262
	PrecMin	813	85	69	92	15	8	1	5	3	5	34	67	97
	PrecStd	236	90	88	56	32	27	33	27	18	46	50	47	53
	DB	35.8	34.8	33.1	31.9	29.2	29.1	29.2	32.2	35.1	35.2	34.0	34.1	
	MCWB	22.0	21.2	22.1	21.0	19.7	18.5	17.5	17.5	18.6	19.8	21.0	21.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	33.0	33.0	31.9	30.2	28.0	27.1	27.9	30.0	32.2	32.9	31.9	32.8	
	2% MCWB	21.9	21.7	21.6	20.9	19.3	18.1	17.0	17.0	18.3	19.9	21.0	21.0	
	5% DB	31.2	31.9	30.5	29.0	26.2	25.8	26.2	28.2	30.1	30.9	30.0	31.0	
	5% MCWB	22.0	21.9	21.7	20.8	18.8	17.6	17.0	17.0	18.1	19.8	20.7	21.1	
	10% DB	29.9	30.1	29.0	27.2	24.8	23.9	24.7	26.2	27.9	28.2	28.1	29.2	
	10% MCWB	22.0	21.9	21.8	20.2	18.5	17.3	16.7	16.8	18.2	19.7	20.4	21.3	
	WB	25.0	24.8	24.4	23.2	21.7	20.4	19.5	20.2	21.7	22.9	23.4	23.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.6	30.1	28.6	28.2	26.1	25.0	24.9	26.7	28.3	28.6	28.5	28.2	
	2% WB	24.0	23.8	23.6	22.5	20.7	19.2	18.6	18.9	20.6	21.9	22.5	23.1	
	2% MCDB	29.0	29.2	28.1	27.1	24.9	23.9	23.9	25.6	26.6	28.3	27.8	28.2	
	5% WB	23.3	23.2	23.0	21.8	19.9	18.6	17.9	18.1	19.9	21.2	21.8	22.6	
	5% MCDB	28.1	28.1	27.5	26.0	24.1	23.3	23.3	24.8	25.7	26.9	26.9	27.6	
	10% WB	22.8	22.7	22.5	21.2	19.1	17.8	17.2	17.5	19.1	20.5	21.3	22.0	
	10% MCDB	27.3	27.2	26.7	25.1	23.0	22.3	22.5	23.6	24.8	25.6	25.9	26.8	
Mean Daily Temperature Range	MDBR	10.0	10.3	9.7	9.9	10.2	11.1	12.1	12.9	11.7	10.8	10.4	10.3	
	5% DB	12.4	12.5	11.6	11.7	12.3	13.7	15.0	16.5	16.6	15.1	13.3	12.7	
	MCWBR	4.1	3.8	3.7	4.0	4.9	5.7	5.8	5.6	5.2	4.7	4.4	4.1	
	5% WB	10.7	10.9	10.2	10.1	11.0	11.3	12.5	13.6	12.8	12.1	11.5	10.8	
	MCWBR	4.2	4.0	4.0	4.0	4.9	5.4	5.4	5.3	5.0	4.7	4.4	4.2	
Clear-Sky Solar Irradiance	taub	0.396	0.393	0.385	0.364	0.352	0.346	0.356	0.391	0.482	0.455	0.430	0.404	
	taud	2.463	2.452	2.477	2.508	2.509	2.512	2.465	2.340	2.073	2.218	2.305	2.415	
	Ebn at Noon	948	938	918	895	865	850	852	854	813	870	910	942	
	Edh at Noon	120	119	111	99	91	87	94	116	164	149	140	126	
All-Sky Solar Radiation	RadAvg	5.40	5.70	5.03	4.54	3.69	3.42	3.59	4.30	4.56	5.07	5.29	5.58	
	RadStd	0.76	0.55	0.47	0.34	0.26	0.30	0.35	0.36	0.56	0.55	0.30	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.30	-0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page